605.486 – Android Development

Module 15
Maps

Google Maps API

- Need Google Play Services
  - Can get through Android SDK Manager (under Extras)
- Need Google Maps API Key
  - Debug key is in <user>/.android/debug.keystore
  - For deployment, use your real key

Get SHA1 of your signing key

- Copy SHA1 fingerprint

```
keytool -list -v -alias
androiddebugkey
        -keystore debug.keystore
        -storepass android
        -keypass android
```

Google APIs Console

- https://code.google.com/apis/console
- APIs & auth
  - APIs -> Google Maps Android API v2 -> ON
  - Registered apps -> Register App
    - Name -> (Your choice)
    - Platform -> Android
      - Accessing APIs directly from Android
  - Past package name, signing fingerprint
  - Get android key

Application Manifest

```
<meta-data
    android:name="com.google.android.maps.v2.API_KEY"
    android:value="key-from-API-console"/>
<uses-feature
        android:glEsVersion="0x00020000"
        android:required="true"/>

<uses-permission
    android:name="android.permission.INTERNET"/>
<uses-permission
    android:name="android.permission.ACCESS_NETWORK_STATE"/>
<uses-permission
    android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
<uses-permission
  android:name=
  "com.google.android.providers.gsf.permission.READ_GSERVICES"/>
```

Example 1 - Basic Map

- Create new Android Project from exist source code
  - Use <android-sdk>/extras/google/google_play_services /libproject/google-play-services_lib
- Add the above as library dependency
- Add MapFragment or MapView to layout

Example 2 - Map Settings

- Can be set in code or XML
  - This example uses XML

Example 3 - Using “My Location”

- Requires new permissions
  - *android.permission.ACCESS_COARSE_LOCATION*
  - *android.permission.ACCESS_FINE_LOCATION*

- Get map from MapFragment
- Enable My Location
- On first update
  - animate camera to its location

Example 4 - Markers

- Get images
- Create marker for location
- Icon Credit
  - http://www.dapino-colada.nl/mini-vector-car-icons
  - License: Creative Commons Attribution-Non-Commercial 3.0

Example 5 - Remembering Locations

- Setup menu items
- When pressed add new marker
- Save in shared prefs

Example 6 – Faking Location

- Long press to set fake location
- Tap to remove fake location
- Long press fake location icon to start drag

Example 7 - Showing Route

- AsyncTask
- Google Directions API (REST/JSON)
- Parse Results
- Create PolyLine on map

Example 8 - Walking / Navigation

- Change route mode to “walking”
- Start google navigation
  - NOTE: Maps API terms state you cannot use their data to perform navigation! We must ask the google app to do it